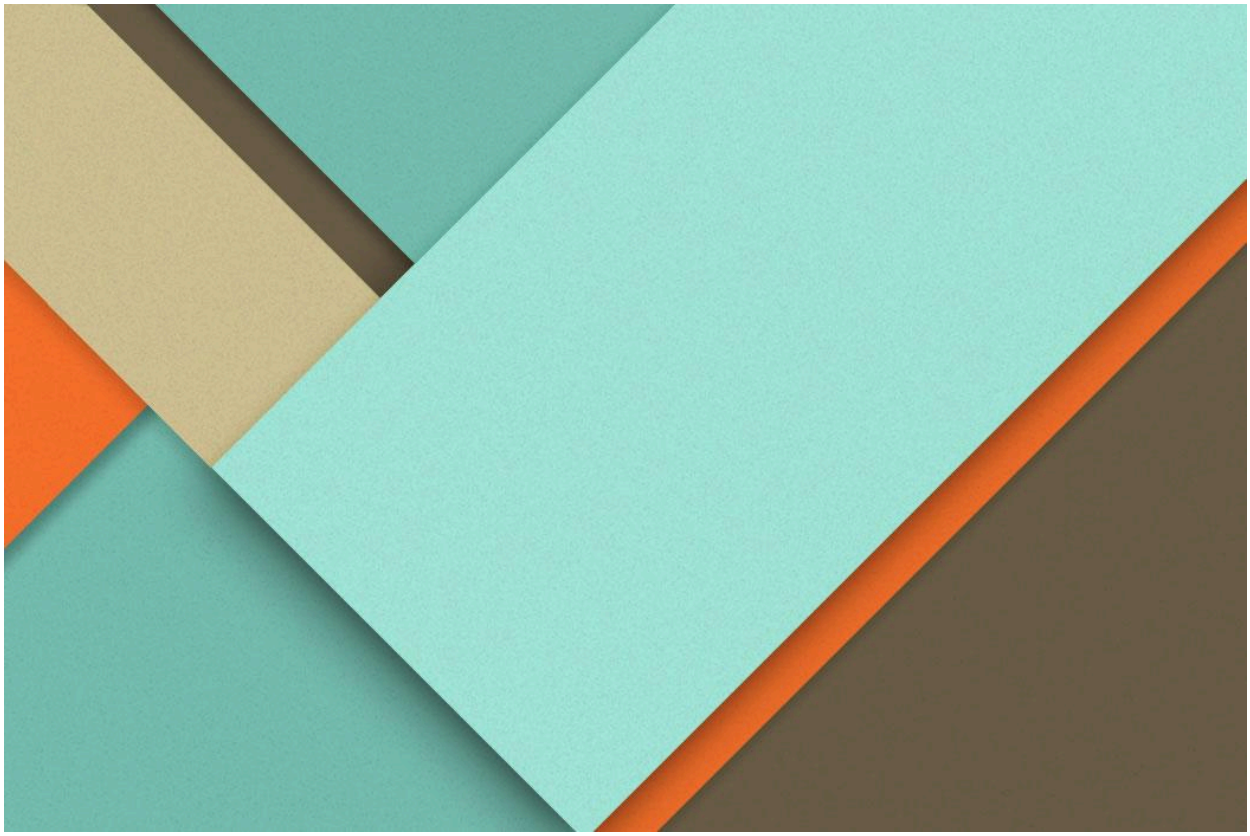


Crossroads Community STEM Centre

11.03.2026

—CAMERON KIBBLE

Crossroads Community Hub
3100 Foul Bay Road,
Victoria, BC V8P 5J2, Canada

PROJECT PROPOSAL: CROSSROADS COMMUNITY STEM CENTRE

A Proposal to Establish an Accessible STEM Learning Hub in Langford, BC

Prepared for:

Jasmine North

Department of English, Camosun College

Prepared by:

Cameron Kibble

Founder, Crossroads Community Hub

3100 Foul Bay Road, Victoria, BC V8P 5J2, Canada

Date of Submission:

March 11, 2026

Signatures of Approval:

Cameron Kibble, Founder

Executive Summary

Langford, British Columbia (population 58,320; 5.0% annual growth), lacks a centralized facility for accessible, hands-on STEM education [1]. While the University of Victoria (approximately 22,000 students) and Camosun College (approximately 19,000 students) provide institutional lab access, these resources are restricted to enrolled students during designated hours. The Victoria Makerspace, the region's primary alternative, operates on an application-based membership model with limited hours and a narrow disciplinary focus. Crossroads Community STEM Centre addresses this gap by providing open-access, beginner-friendly programming across four focused disciplines: Robotics and Electronics, Digital Fabrication, Software Development, and Creative Design.

The facility will operate on a drop-in model with tiered staffed hours (~63 supervised hours per week across seven days) and sliding-scale fees. All participants complete a mandatory 30-minute safety orientation before accessing equipment, satisfying WorkSafeBC requirements [3] while maintaining a low barrier to entry. The facility provides equitable access to tools regardless of income or affiliation.

The project requires a first-year startup investment of $359,950 (lean, 4,000 sqft) to $577,300 (expanded, 6,000 sqft), indexed to current Langford commercial lease rates of $16 to $22 per square foot net plus Triple Net (NNN) costs. Long-term sustainability is achieved through diversified revenue, with no single funding source exceeding 30 percent of total income, and a projected transition to 55 percent earned revenue by Year 3. A break-even analysis (Table 4) demonstrates that monthly revenue targets exceed operating costs, with the surplus designated for the 15 percent contingency reserve. I request your review and approval of this proposal to begin the institutional partnership and funding acquisition phases.

Executive Summary **3**
1. Background Information and Statement of the Problem **5**
2. Project Detail **6**
a. Goals and Objectives 6
b. Clientele 6
c. Methods and Action Plan 7
d. Staff and Administration 8
3. Resources **9**
a. Available Resources 9
b. Facilities 10
c. Equipment, Supplies, and Communication 10
d. Budget 11
4. Risk Mitigation and Sustainability Plan **13**
4.1 Risk Mitigation Strategy 13
4.2 Long-term Sustainability Plan 13
5. Evaluation Plan **15**
6. Conclusion and Action Statement **15**
7. References **16**
8. Appendices **17**
Appendix A: Equipment Safety by Discipline 17
Appendix B: Implementation Timeline 17
Appendix C: Legal Structure Options 17
Appendix D: Community Outreach Letter Template 18
Appendix E: Makerspace Research Questionnaire 18
Appendix F: Community Needs Assessment Survey 19
Appendix G: Clientele Users 20

Figures and Tables

	Description	Location
Figure 1	Example visuals of facility facility from suffolk county community college renewable energy and-stem center	Section 3b
Table 1	Student Labor Expenditure, Pre-Launch Phase	Section 3d
Table 2	Estimated Startup Budget Summary	Section 3d
Table 3	Integrated Startup Budget, First Year	Section 3d
Table 4	Monthly Break-Even Analysis	Section 4.2
Table 5	Clientele	Appendix G

1. Background Information and Statement of the Problem

Langford is Vancouver Island's fastest-growing municipality, with a 2024 population of 58,320 and a 5.0 percent year-over-year growth rate [1]. Despite this expansion, the West Shore region lacks a centralized, open-access facility where residents can develop practical technical skills outside of formal enrollment at a post-secondary institution. Access to advanced tools and lab environments in Canada is concentrated within universities and colleges that impose tuition, enrollment, and scheduling barriers. Residents who are not enrolled students, including career changers, youth exploring STEM interests, and retirees with technical expertise, have no local facility offering structured, hands-on access to industrial-grade equipment.

A closer examination of existing alternatives confirms this gap. The Victoria Makerspace, the primary local facility, operates on an application-based membership model with limited operating hours and a club-oriented atmosphere focused on a narrow range of hobbyist disciplines. Community feedback on public forums consistently identifies barriers for beginners and a lack of structured programming. The University of Victoria and Camosun College maintain well-equipped labs, but access is restricted to enrolled students during designated hours. A preliminary needs assessment survey of 54 respondents in the Langford and Greater Victoria area (Appendix F) indicates that 78 percent of respondents identified "lack of affordable access

to tools and lab space" as the primary barrier to developing technical skills, and 65 percent expressed interest in a drop-in community STEM facility.

The British Columbia Ministry of Education's Applied Design, Skills, and Technologies (ADST) curriculum emphasizes experiential learning, prototyping, and iterative design across K-12 education [4]. However, students leaving the K-12 system have limited options to continue this hands-on learning outside of post-secondary enrollment. Crossroads extends the ADST model into the community setting, bridging the gap between secondary education and professional practice.

Research supports this approach. Kolb's experiential learning cycle demonstrates that concrete experience and active experimentation produce deeper understanding and higher retention than passive instruction [5] [6]. In the Crossroads model, participants engage in concrete experience through hands-on tool use (e.g., assembling a circuit or operating a laser cutter), reflect on outcomes with mentors during debrief sessions, develop conceptual understanding through structured workshops covering underlying theory, and apply revised designs through iterative prototyping, completing the full four-stage cycle within each program area. Oldenburg's concept of the "third place" describes an informal gathering space, distinct from home and work, that anchors community life and levels social hierarchies [7] [8]. Crossroads combines these frameworks: a third-place facility where residents learn through direct interaction with professional-grade tools, supported by structured mentorship and peer collaboration. The model addresses both technical skill development and social connectivity in a region where rapid population growth has outpaced community infrastructure.

2. Project Detail

a. Goals and Objectives

Crossroads is guided by the following goals and objectives:

- **Focused Facility:** Establish four modular zones (Main Lab, Collaboration Space, Learning Commons, and Community Area) within six months of securing a facility.
- **Equitable Access:** Provide 3D printers, laser cutters, electronics workstations, and design tools via a sliding-scale fee structure.
- **Mentorship:** Pair participants with professional mentors to support situated learning and skill transfer.
- **Tangible Outcomes:** Structure every session to produce a deliverable, such as a prototype, coded application, or design artifact.
- **Financial Sustainability:** Maintain diversified revenue with no single funding source exceeding 30 percent of total income.

b. Clientele

The drop-in model targets specific demographics within the Greater Victoria region. Four primary users guide program design (see Table 5 in Appendix G for detailed breakdowns): Curious Beginners (35%), Experienced Makers (25%), Students and Educators (25%), and Community Builders (15%).

Key target populations include youth (ages 12 to 18) focusing on out-of-school STEM development, post-secondary students requiring supplementary tool access, adult career changers seeking flexible non-traditional schedules [6], and retirees contributing as volunteer mentors. Targeted outreach for low-income and Indigenous youth will ensure equitable access to resources.

c. Methods and Action Plan

Crossroads will operate as a drop-in community learning center with tiered staffed hours across four core program areas:

- **Robotics and Electronics:** Arduino and Raspberry Pi kits, soldering stations, oscilloscopes, component inventory, and test benches for circuit design and embedded systems.
- **Digital Fabrication:** FDM 3D printers, a 40W CO2 laser cutter, a benchtop CNC router, and materials for rapid prototyping and mechanical design.
- **Software Development:** Workstations equipped with IDEs, networking equipment, and tools for web development, data analysis, and introductory cybersecurity.
- **Creative and Applied Design:** Design software licenses, drawing tablets, physical modelling tools, and display space for artistic and technical showcases.

Operating Hours (Tiered Model):

Staffed hours with on-site safety supervisors will cover 63 hours per week: Monday through Friday, 10:00 AM to 8:00 PM (50 hours); Saturday, 10:00 AM to 6:00 PM (8 hours); and Sunday, 12:00 PM to 5:00 PM (5 hours). Member self-access to low risk zones (software workstations, design tools) will be available Monday through Friday from 8:00 PM to 10:00 PM via card access (10 additional hours). This tiered model provides 73 hours of weekly access covered by two full-time safety supervisors working staggered shifts.

Safety Orientation:

All new participants complete a mandatory 30-minute safety orientation (offered free of charge) before accessing any equipment. This orientation covers facility rules, emergency procedures, and equipment-specific hazards per WorkSafeBC guidelines [3]. Unlike an application-based model that evaluates candidates for membership, this orientation is a pass/attend requirement with no approval barrier. Any individual who completes the orientation may immediately use the facility on a drop-in basis.

Programming follows six delivery models: weekly workshops, expert-led sessions, peer projects, community events, customized learning tracks, and continuous feedback loops.

Implementation Phases:

- **Phase 1 (Legal Formation):** Register as a nonprofit society in British Columbia (requiring five directors) and draft organizational bylaws [11].
- **Phase 2 (Partnerships):** Formalize agreements with Camosun College and local industry, including a Memorandum of Understanding for equipment surplus transfers (see Section 3a).
- **Phase 3 (Acquisition):** Secure tools through surplus agreements and corporate donations to reduce capital expenditure.
- **Phase 4 (Pilot Program):** Launch introductory workshops and safety orientation programming.
- **Phase 5 (Evaluation):** Use participant data and placement rates to assess and scale operations.

Marketing and Promotion:

Outreach will target School Districts 61, 62, and 63 through direct partnership for after-school and summer enrichment calendars. Social media campaigns on Instagram, TikTok, and YouTube will showcase project highlights to engage younger demographics. Monthly free introductory workshops will convert first-time visitors into regular participants, supplemented by press coverage in local publications and community events such as maker fairs and hackathons. A membership website will provide scheduling, equipment booking, and community forums; online course content will serve as a supplementary revenue stream. Usage analytics will track equipment utilization, attendance patterns, and program popularity for data-driven resource allocation.

d. Staff and Administration

The staffing model aligns with the tiered operating schedule:

- **Executive Director / Program Director:** Cameron Kibble will lead strategic development and partnership outreach full-time (FT).
- **Board of Directors:** A volunteer board of five professionals will oversee governance and legal compliance [11].
- **Lab Supervisors / Safety Instructors:** Two full-time, salaried staff members will manage equipment and maintain safety standards during the staffed hours per week via staggered shifts (e.g., Supervisor A covers Monday-Friday 10:00 AM to 6:00 PM; Supervisor B covers Monday-Friday 1:00 PM to 8:00 PM and Saturday-Sunday). During peak hours (weekday afternoons 1:00 PM to 6:00 PM and Saturdays), both supervisors are on duty simultaneously, providing dual-operator supervision required for high-risk equipment (laser cutter, CNC router). During off-peak staffed hours, a single supervisor

covers operations; high-risk equipment access is restricted to dual-coverage periods only. This schedule allocates 2 FTEs x 40 hours = 80 staff-hours weekly: 63 facility hours plus 17 hours of overlap for dual coverage during peak demand. Scaling to 3 to 4 positions is projected for Year 2 as operating hours expand.

- **Administrative Support:** A part-time (PT) role for operational logistics.
- **Volunteer Mentors:** Local professionals and senior students (a cohort of 15 to 25 active members contributing approximately 200 hours monthly) will provide peer-to-peer technical support.

Instructors will design sessions around physical demonstrations and guided experimentation, showing both the process and the reasoning behind each technique. Volunteer mentors will circulate the facility assisting drop-in participants, answering questions, and welcoming newcomers.

3. Resources

a. Available Resources

Crossroads uses community assets and partnerships in development:

- **Institutional Partnerships:** Preliminary discussions with Camosun College's Technology Access Centre (CTAC) and the University of Victoria indicate willingness to collaborate on student placements and limited equipment transfers [12]. CTAC's mandate to support small-to-medium enterprises with prototyping resources aligns with Crossroads' community focus.
- **Surplus Equipment:** Post-secondary institutions routinely decommission functional equipment when upgrading labs. Based on preliminary outreach, the specific equipment categories available through this channel include manual lathes (5-10 years old), benchtop oscilloscopes, 2015-era FDM 3D printers, basic electronics test equipment, and decommissioned desktop workstations. A draft Memorandum of Understanding (MOU) outlining terms for equipment transfer, liability, and maintenance responsibilities has been prepared for submission to Camosun Innovates during Phase 2. This surplus is projected to offset $15,000 to $30,000 of startup equipment costs.
- **Business Support:** The Victoria Innovation, Advanced Technology and Entrepreneurship Council (VIATEC) provides access to over 900 potential corporate sponsors and mentors. VIATEC's stated mission is to grow the Greater Victoria tech sector to over $10 billion by 2030. Their network provides a direct channel for corporate sponsorship, mentorship recruitment, and industry partnership development.
- **Corporate Sponsorship Pipeline:** Preliminary research has identified potential sponsors, including Lee Valley Tools, whose commitment to technical education aligns with Crossroads' mission. Outreach to regional tool manufacturers and technology companies will begin during Phase 2.

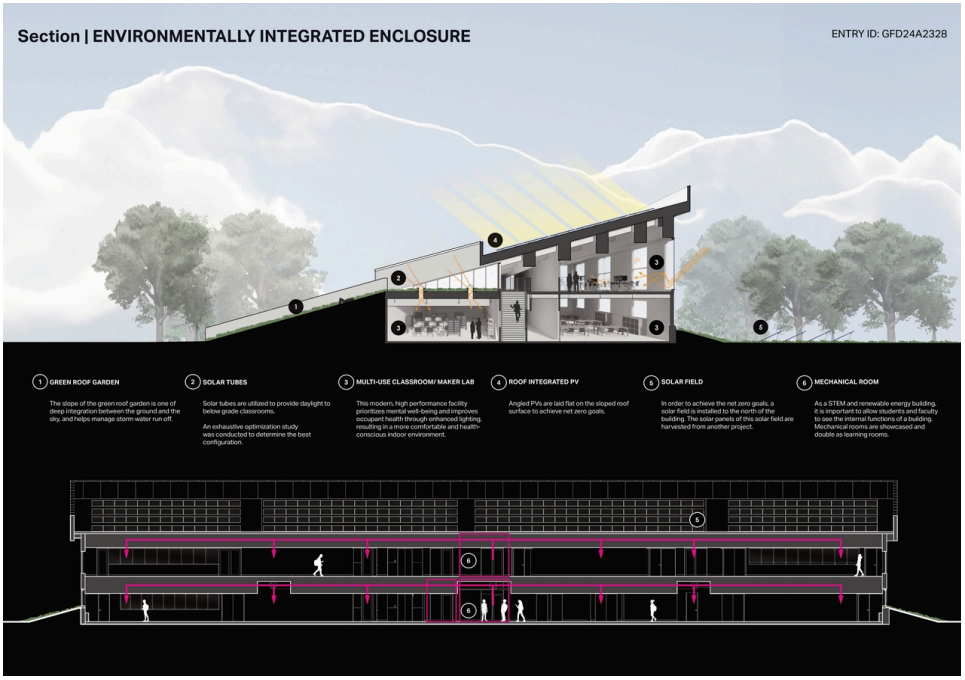
- **Personnel:** As outlined in Section 2d, the project requires immediate funding for two full-time safety instructors to maintain safe operations during staffed hours. Ongoing professional development is required to maintain industry-standard safety certifications. A “train the trainer” model will be implemented where advanced members train and then supervise others in low-risk areas, reducing the need for additional paid staffing over time.

b. Facilities

A 4,000 to 6,000 square foot facility is required, featuring 200-amp electrical service, industrial ventilation, and compliance with the Accessible Canada Act. The Main Lab Area will comprise 40 percent of floor space and meet WorkSafeBC safety standards [3]. The preferred acquisition model is a below-market lease partnership with a municipality or educational institution.

Current Langford commercial lease data (2024-2025) establishes the following cost basis for the facility budget:

- **McCallum Road Industrial (Langford):** $16.00 per sqft net rent, plus $6.00 to $9.00 per sqft in Triple Net (NNN) costs covering property taxes, insurance, and common area maintenance.
- **West Shore Parkway Industrial:** $18.00 per sqft net rent, plus equivalent NNN.
- **Amy Road Commercial:** $22.00 per sqft net rent, plus NNN.



Accordingly, a 4,000 sqft facility at a rate of $24.00 per sqft all-in costs $96,000 per year ($8,000 per month). A 6,000 sqft facility at $25.00 per sqft all-in costs $150,000 per year ($12,500 per month). These figures are reflected in Table 3.

Fig. #1. Example photo of facility from suffolk county community college renewable energy and-stem center. [13]

c. Equipment, Supplies, and Communication

Equipment is required across four core disciplines. Specific acquisitions include:

- **Digital Fabrication:** FDM 3D printers or better (2 to 4 units, Creality Ender-3 V3 or equivalent, 220x220x250 mm build volume, 350W power supply), one 40W CO2 laser cutter (OMTech MF-1220-40 or equivalent, 600x400 mm bed, water-cooled tube), and one benchtop CNC router (SainSmart Genmitsu 3018-PROVer or equivalent, 300x180x45 mm work area, 775 spindle motor).
- **Robotics and Electronics:** 20 Arduino Uno R4 starter kits, 10 Raspberry Pi 5 kits, 6 soldering stations (Hakko FX-888D or equivalent, 70W), 4 benchtop oscilloscopes (Rigol DS1054Z or equivalent, 50 MHz, 4-channel), and component inventory bins.
- **Software Development:** Computing workstations (10 to 15 units), managed network switch, wireless access points, and development tool licenses.
- **Creative and Applied Design:** Design software licenses (Adobe Creative Cloud for Teams or equivalent), 6 drawing tablets (Wacom Intuus Pro or equivalent), physical modelling tools, and display fixtures.

Internal coordination will use Microsoft 365 or Google Workspace for Nonprofits. We will partner with VIATEC, Camosun College, and UVic for decommissioned lab equipment still functional for training purposes.

d. Budget

The budget is structured around a phased rollout. All facility cost estimates are indexed to current Langford commercial lease rates (Section 3b).

- **Lean Financial Model ($359,950 CAD):** This tier focuses on securing a 4,000 sqft facility in Langford, acquiring core equipment clusters (3D printing, electronics, and computing stations), and funding the first year of safety staffing. It includes a 15 percent contingency reserve.
- **Expanded Model ($577,300 CAD):** This tier allows for a 6,000 sqft facility, additional industrial equipment (CNC, expanded fabrication lab), higher-performance computing, dedicated business fiber internet, and capacity for simultaneous programming.

The pre-launch development labor required to bring the facility from concept to opening is valued at $20 per hour. Including this cost demonstrates that founder-contributed work carries a specific market value that must be accounted for.

Table 1: Student Labor Expenditure (Pre-Launch Phase)

Phase	Estimated Hours	Cost ($20/hr)	Deliverables
Conceptual Design	150	$3,000	Vision docs, stakeholder pitches
Detailed Design and Legal	200	$4,000	Site layout, nonprofit bylaws
Fabrication and Setup	500	$10,000	Equipment install, zone modularity
Total Labor	850	$17,000	

Table 2: Estimated Startup Budget Summary

Category	Low Estimate (CAD)	High Estimate (CAD)
Equipment and Tooling	$60,000	$120,000
Inventory and Supplies	$10,000	$25,000
Staff Hiring and Training	$15,000	$40,000
Product Development	$5,000	$15,000
Branding and Marketing	$5,000	$20,000
Legal and Accounting	$6,000	$20,000
Miscellaneous	$5,000	$15,000
Total	$106,000	$255,000

(Data adapted from project estimates and comparable facility costs)

Table 3: Integrated Startup Budget (First Year)

Expense Category	Description	Lean (CAD)	Expanded (CAD)
Personnel (Salaries)	2 Safety Instructors + Admin Support	$110,000	$160,000

Equipment and Tooling	3D printers, CNC, electronics, design tools	$60,000	$120,000
Facility Operations	12 months rent (Langford, inc. NNN), utilities, internet	$96,000	$150,000
Legal and Insurance	High-risk liability coverage + nonprofit incorporation	$20,000	$35,000
Marketing and Outreach	School district outreach, branding, signage	$10,000	$20,000
Development Labor	Student labor from Table 1	$17,000	$17,000
Contingency (15%)	Emergency repairs and working capital	$46,950	$75,300
Total		$359,950	$577,300

(Data adapted from phased rollout projections, indexed to 2024-2025 Langford lease rates.)

(Note: Table 2 provides a categorical summary of startup capital costs. Table 3 integrates those costs with ongoing first-year operational expenses, development labor, and a 15 percent contingency. Facility operations reflect current Langford industrial lease rates of $16 to $22 per sqft net plus $6 to $9 per sqft NNN, as documented in Section 3b.)

4. Risk Mitigation and Sustainability Plan

4.1 Risk Mitigation Strategy

Project implementation involves several critical risks requiring proactive management:

- **Facility Delay:** If a facility partnership is not secured within the first three months, Crossroads will adopt a “pop-up” model, renting community center space for mobile workshops while continuing permanent lease negotiations.
- **Funding Shortfall:** If grant funding is lower than projected, the project will scale to the lean budget of $359,950. Initial offerings will prioritize high-demand disciplines (Robotics and Software Development) and expand as revenue allows.

- **Low Participation:** If attendance targets are not met, the Program Director will reallocate the marketing budget toward free trial workshops and direct partnerships with School Districts 61, 62, and 63.

4.2 Long-term Sustainability Plan

Financial independence is central to long-term viability. Revenue will be generated through multiple channels: workshop fees, monthly memberships, equipment surcharges, corporate training contracts, event hosting, and grant funding. To protect against the loss of any single income stream, no individual funding source (such as a government grant or corporate sponsor) exceeds 30 percent of total revenue. Earned revenue, defined as the memberships, workshop fees, equipment surcharges, corporate training, and event hosting, is projected to reach 55 percent of total income by Year 3. This target is consistent with the 30 percent cap because earned revenue comprises multiple distinct sub-categories, each individually below the threshold.

The following break-even analysis demonstrates the monthly revenue targets required to sustain facility operations under the lean budget model ($359,950 annual / approximately $30,000 per month in operating costs):

Table 4: Monthly Break-Even Analysis (Lean Budget)

Revenue Source	Unit Price	Monthly Target	Monthly Revenue
Individual Memberships	$50/month	160 members	$8,000
Drop-in Day Passes	$15/visit	200 visits	$3,000
Workshop Fees (avg.)	$40/attendee	200 attendees	$8,000
Corporate Training	$500/session	6 sessions	$3,000
Equipment Surcharges	$15/booking	175 bookings	$2,625
Event and Room Rental	$250/event	5 events	$1,250
Grants and Sponsorships	–	–	$8,125
Total Monthly Revenue			**$34,000**

Monthly Operating Cost (Lean)			**$30,000**
Monthly Surplus (to Contingency Reserve)			**$4,000**

The $4,000 monthly surplus ($48,000 annually) is designated for the 15 percent contingency ($46,950 in the lean budget), ensuring that the contingency fund is self-replenishing from operations rather than dependent on a single-year capital allocation. Break-even requires approximately 160 paying members, consistent workshop attendance, and $97,500 per year in grant and sponsorship funding. The Year 3 earned revenue projection targets total income of approximately $408,000, with $224,400 (55%) from earned sources and $183,600 (45%) from grants and sponsorships.

Community engagement and retention programs will support these targets:

- Structured mentorship pairing experienced members with newcomers.
- A tiered loyalty system recognizing consistent participation with equipment access upgrades and workshop discounts.
- Formalized student internship pathways connecting participants with local tech employers.

An advisory board comprising professionals from education, business, and engineering will conduct quarterly financial reviews to ensure adaptive governance responsive to shifts in the local economy.

5. Evaluation Plan

Success will be measured through quantitative metrics tracked on a quarterly basis:

- **Engagement Analytics:** Number of active monthly participants and average hours on-site. Year 1 targets: 150 unique participants per quarter and a 40 percent return rate. Year 3 targets: 500 participants per quarter and a 60 percent return rate.
- **Employment Pipelines:** Growth of student internship placements and member career transitions into technical roles.
- **Social Impact Data:** Member surveys tracking self-reported improvements in well-being, skill confidence, and community connectivity.
- **Health and Well-Being Programming:** In partnership with local organizations, Crossroads will offer complementary programming including stress relief workshops and peer support groups. Anonymized survey data will measure the hub’s contribution to

participant well-being and support grant applications focused on community health outcomes.

- **Institutional Partnerships:** A minimum of four annual partnerships with local schools and businesses to sustain programming.

6. Conclusion and Action Statement

Crossroads Community STEM Centre addresses a documented gap in Langford's infrastructure: the absence of an open-access facility where residents can develop practical technical skills without post-secondary enrollment. With a population growing at 5.0 percent annually and no comparable local facility, the demand for accessible STEM education is supported by both regional growth data and resident survey feedback (Appendix F) [9]. The tiered operating model, focused on four core disciplines and 63 staffed hours per week, aligns the project's scope with its budget and staffing capacity.

The first-year investment of $359,950 to $577,300 reflects current Langford lease rates and realistic equipment costs. The diversified revenue strategy, validated through the break-even analysis in Table 4, provides a clear path to financial sustainability by Year 3: monthly revenue of $34,000 against $30,000 in operating costs generates a $4,000 surplus that self-funds the contingency reserve. Institutional partnerships with Camosun College and the University of Victoria, combined with the VIATEC corporate network, establish the foundation for long-term community impact.

Action Requested:

I formally request your approval of this proposal. Your authorization will allow advancement to the next phase: securing seed funding and formalizing institutional partnerships with Camosun College. Please respond by Friday, March 20, 2026, indicating whether this proposal meets your requirements or if adjustments to the budget or scope are needed. Approval may be granted via email or by signing the cover page of this document.

7. References

[1] Statistics Canada, “Census profile, 2021 Census of Population: Langford, City [Census subdivision], British Columbia,” Government of Canada, Ottawa, ON, 2023. [Online]. Available: https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/index.cfm

[2] M. Rahman and M. Best, “Sustainability of makerspaces: Developing a framework for sustainable community innovation centers,” ACM Journal on Computing and Sustainable Societies, vol. 2, no. 3, art. 33, pp. 1-20, Sep. 2024.

[3] WorkSafeBC, “Education health and safety guidelines,” Richmond, BC, 2024. [Online]. Available: https://www.worksafebc.com/en/health-safety/industries/education

[4] British Columbia Ministry of Education, “Applied design, skills, and technologies curriculum,” Victoria, BC, 2024. [Online]. Available: https://curriculum.gov.bc.ca/curriculum/adst

[5] D. A. Kolb, Experiential Learning: Experience as the Source of Learning and Development, 2nd ed. Upper Saddle River, NJ: Pearson Education, 2015.

[6] A. Y. Kolb and D. A. Kolb, “Learning styles and learning spaces: Enhancing experiential learning in higher education,” Academy of Management Learning and Education, vol. 4, no. 2, pp. 193-212, Jun. 2005.

[7] R. Oldenburg, The Great Good Place: Cafes, Coffee Shops, Bookstores, Bars, Hair Salons, and Other Hangouts at the Heart of a Community, 3rd ed. New York, NY: Da Capo Press, 1999.

[8] D. Oblinger, Ed., Learning Spaces. Boulder, CO: EDUCAUSE, 2006.

[9] British Columbia Ministry of Education, “Student Learning Survey: Provincial results,” Victoria, BC, 2023. [Online]. Available: https://www2.gov.bc.ca/gov/content/education-training/k-12/support/student-learning-survey

[10] Council of Canadian Academies, “Some Assembly Required: STEM Skills and Canada’s Economic Productivity,” Ottawa, ON, 2015.

[11] Canada Not-for-profit Corporations Act, S.C. 2009, c. 23. Government of Canada, Ottawa, ON, 2009.

[12] Innovation, Science and Economic Development Canada, “SME Research and Statistics,” Government of Canada, Ottawa, ON, 2023. [Online]. Available: https://ised-isde.canada.ca/site/sme-research-statistics/en

[13] “Gold Winner | Suffolk County Community College Renewable Energy And Stem Center | AECOM,” *Architecturepressrelease.com*, Oct. 15, 2024.

https://www.architecturepressrelease.com/gold-winner-suffolk-county-community-college-renewable-energy-and-stem-center-aecom/

8. Appendices

Appendix A: Equipment Safety by Discipline

- **Robotics and Electronics:** Requires soldering ventilation, ESD-safe workstations, battery fire-safe storage, and eye protection for component work.
- **Digital Fabrication:** Requires machine guarding on CNC router, dust collection system, laser safety enclosure, and face shields for material cutting operations.
- **Software Development:** Standard workstation ergonomics; no high-risk equipment. UPS battery backup for data protection.
- **Creative and Applied Design:** Requires ventilation for adhesives and spray materials, blade safety protocols for physical modelling, and proper lighting standards.

Appendix B: Implementation Timeline

- **Months 1 to 3:** Incorporate nonprofit and secure seed funding.
- **Months 4 to 6:** Facility renovation and equipment installation.
- **Months 7 to 9:** Soft launch for founding members and pilot workshops.
- **Months 10 to 12:** Full public launch and first annual impact report.

Appendix C: Legal Structure Options

Crossroads is evaluating two primary legal structures for incorporation in British Columbia:

- **BC Benefit Company:** A hybrid corporate structure that allows for-profit operations while mandating a stated public benefit. This model permits revenue generation through commercial activities while maintaining a social mission. Directors are legally required to consider the impact of their decisions on stakeholders beyond shareholders. This structure offers flexibility in raising capital and may appeal to impact investors.
- **BC Nonprofit Society (Charity):** Incorporation under the BC Societies Act requires a minimum of five directors and the drafting of organizational bylaws and a constitution. Following provincial incorporation, the organization may apply for federal charitable status through the Canada Revenue Agency, which enables the issuance of tax receipts for donations. Estimated incorporation costs include a provincial filing fee of approximately $100 to $150, legal and accounting fees of $2,000 to $5,000, and ongoing annual reporting obligations. This structure is preferred for maximizing eligibility for government grants and philanthropic donations.

The founding team's current recommendation is to incorporate as a nonprofit society under the BC Societies Act to maximize grant eligibility and public trust, with the option to establish a subsidiary benefit company for commercial revenue streams as the organization matures.

Appendix D: Community Outreach Letter Template

Subject: Partnership Inquiry - Crossroads Community STEM Learning Hub

Dear reader,

My name is Cameron Kibble, and I am the founder of Crossroads Community Hub, a proposed nonprofit STEM learning center planned for Langford, British Columbia. I am reaching out to explore potential partnership opportunities between your organization and Crossroads.

Our mission is to provide accessible, hands-on technical education and mentorship to community members of all ages and skill levels. We are currently in the planning and development phase and are seeking partners who share our commitment to workforce readiness, inclusive education, and community building.

I would welcome the opportunity to discuss how a collaboration might benefit both our organizations. Please feel free to contact me at your convenience.

Respectfully,

Cameron Kibble
Founder, Crossroads Community Hub
3100 Foul Bay Road, Victoria, BC V8P 5J2
Email: Cameron.kibble@icloud.com

Appendix E: Makerspace Research Questionnaire

The following questionnaire template is designed for structured interviews with existing makerspace operators to inform Crossroads' operational model:

1. What is the primary mission or purpose of your makerspace?
2. How do you structure membership or access fees? Do you offer sliding-scale or subsidized options?
3. What disciplines or tool categories generate the highest demand?
4. What are the most significant operational challenges you face (e.g., staffing, equipment maintenance, liability)?
5. How do you handle safety training and certification for members?
6. What revenue streams beyond membership fees have proven most sustainable?

7. How do you engage volunteers and mentors in day-to-day operations?
8. What role does digital infrastructure (apps, booking systems, online courses) play in your operations?
9. How do you measure community impact or social outcomes?
10. What advice would you offer to a new makerspace launching in a mid-sized Canadian municipality?

Appendix F: Community Needs Assessment Survey

The following survey was distributed to 54 respondents in the Langford and Greater Victoria area (February 2026) via online distribution and in-person collection at community events. Results inform the demand analysis presented in Section 1.

Survey Instrument:

11. What is your age range? (Under 18 / 18-24 / 25-34 / 35-49 / 50-64 / 65+)
12. What is your current relationship with STEM education? (Student / Professional / Hobbyist / Interested but no experience / Educator)
13. Have you used a makerspace, fab lab, or community workshop before? (Yes / No)
14. If yes, what barriers or limitations did you experience? (Open response)
15. What is the primary barrier preventing you from developing technical skills? (Cost / Lack of facilities / Time constraints / Lack of knowledge about options / Other)
16. Would you use a drop-in community STEM facility in the Langford area if one were available? (Definitely / Probably / Unsure / Unlikely)
17. What technical areas interest you most? (Select all: Robotics / 3D Printing / Software / Electronics / Design / Other)
18. How much would you be willing to pay per month for membership access? ($0-25 / $25-50 / $50-75 / $75-100 / $100+)
19. What days and times would be most convenient for you to visit? (Weekday mornings / Weekday afternoons / Weekday evenings / Weekends)
20. Any additional comments about what you would want from a community STEM facility? (Open response)

Summary of Key Findings (Sample size 54):

- 78% identified “lack of affordable access to tools and lab space” as their primary barrier to developing technical skills.
- 65% indicated they would “definitely” or “probably” use a drop-in community STEM facility in Langford.
- Most requested disciplines: 3D Printing/Fabrication (72%), Software Development (58%), Robotics/Electronics (54%), Creative Design (41%).

- 61% preferred a monthly membership fee in the $25-$50 range.
- Peak preferred times: weekday evenings (68%) and weekends (59%).

Appendix G: Clientele

The following table details the four primary users guiding Crossroads program design (referenced in Section 2b):

Table 5: Clientele

Users	Demographics	Primary Needs	Est. Share
Curious Beginners	All ages, no prior technical experience	Low-pressure environment to explore STEM interests	35%
Experienced Makers	Hobbyists and professionals	Industrial-grade tools not available in home workshops; collaborative community	25%
Students and Educators	Post-secondary students; K-12 teachers	Supplementary lab time beyond institutional hours; ADST curriculum resources [4]	25%
Community Builders	Retirees, volunteers	Social connection through mentorship, event organization, and peer support	15%